Scalable Sentiment Analysis on Big Data: A Hybrid Architecture Combining Apache Spark, Kafka Streaming, Deep Learning, and Explainable AI

IEEE Technical Project Report • HMCE02T: Scalable Machine Learning & Big Data Analytics

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***Abstract*—With the exponential surge in user-generated online media reviews, traditional single-node Natural Language Processing (NLP) workflows encounter severe computational bottlenecks in scalability, stream ingestion, and inference latency. This paper introduces a scalable, end-to-end sentiment analysis framework combining distributed batch processing, real-time event streaming, deep sequence modeling, and Explainable Artificial Intelligence (XAI). Using Apache Spark (PySpark MLlib) on the 50,000 IMDb movie review benchmark, distributed data cleaning, Tokenization, StopWords filtering, and TF-IDF feature extraction pipelines were constructed to train distributed Logistic Regression (81.81% accuracy, 15.88s training) and Multinomial Naive Bayes (82.33% accuracy, 8.78s training). To address real-time high-throughput demands, an Apache Kafka messaging broker is integrated with Spark Structured Streaming to score incoming textual streams with sub-second latency using pre-fitted pipeline models. For nuanced semantic representations, a deep Long Short-Term Memory (LSTM) network achieving 87.00% accuracy was deployed, complemented by pretrained multilingual Transformer models (RoBERTa and XLM-RoBERTa) for cross-lingual generalization. Crucially, the black-box nature of neural decisions is demystified via Local Interpretable Model-agnostic Explanations (LIME) and an automated LLM Film Critic narrative engine. The complete solution is exposed through a production-grade FastAPI microservice and an interactive web interface ("Reel Verdict").

***Index Terms*—Big Data Analytics, Apache Spark, PySpark MLlib, Apache Kafka, Spark Structured Streaming, Sentiment Analysis, LSTM, RoBERTa, Explainable AI (XAI), LIME, FastAPI.**

I. Detailed Problem Statement

A. Context and Motivation

In modern digital commerce, streaming entertainment platforms, and social networks, opinion mining and sentiment classification represent vital pillars for recommendation engines, automated brand reputation monitoring, and business intelligence. Modern online portals receive hundreds of thousands of user critiques daily. This empirical domain epitomizes the core challenges of Big Data:

- **Volume:** Large-scale textual corpora exceed memory limits on standard workstations, precipitating Out-Of-Memory (OOM) failures during high-dimensional n-gram tokenization and matrix decomposition.
- **Velocity:** Reviews arrive continuously in real time. Batch architectures that process data periodically introduce hours of latency lag, precluding immediate response to viral events, PR crises, or live audience reactions.
- **Variety:** Reviews comprise unstructured colloquial prose, colloquial slang, typos, HTML tags, and multilingual text (e.g., Hindi, Spanish, French, Arabic).

B. Inadequacies of Existing Systems

Conventional implementations rely on single-node libraries (such as scikit-learn or single-threaded pandas) that operate within a single process memory space and fail to leverage multi-node distributed clusters or partitioned Resilient Distributed Datasets (RDDs). Furthermore, model deployment often suffers from train-serve skew when batch pipelines cannot be directly re-used for real-time streaming data. Lastly, deep neural networks function as opaque black boxes, lacking interpretability and algorithmic accountability.

C. Course Outcomes Alignment

This project systematically targets the requirements of **HMCE02T**:

- **CO2 (Distributed vs. Traditional Trade-offs):** Quantify empirical performance, training latency, scalability, and resource footprints of distributed classifiers (Spark MLlib) versus single-node deep neural networks.
- **CO3 (PySpark Pipelines):** Engineer unified, modular, reproducible PySpark Pipelines encapsulating distributed text sanitization, tokenization, stop-word elimination, TF-IDF vectorization, and model fitting.
- **CO4 (Streaming vs. Batch Analytics):** Build dual-mode operational workflows demonstrating distributed batch analytics over static data alongside real-time event-driven micro-batch stream scoring via Apache Kafka and Spark Structured Streaming.

II. Dataset Description

The project evaluates the canonical **IMDb Large Movie Review Dataset**:

- **Total Instances:** 50,000 labeled movie reviews.
- **Class Balance:** Perfectly balanced binary dataset comprising 25,000 positive reviews ($\geq$ 7/10 rating) and 25,000 negative reviews ($\leq$ 4/10 rating).
- **Storage Footprint:** ~66.2 MB raw uncompressed CSV.
- **Attributes:** `review` (raw text string) and `sentiment` (ground truth label).

Distributed data cleaning implemented in PySpark includes regex stripping of HTML break tags (`<br />`), filtering non-alphabetical characters, lowercasing, whitespace trimming, and duplicate elimination. The cleaned corpus is persisted into columnar Apache Parquet format (`clean_reviews.parquet`) leveraging Snappy

compression, and split into an 80% training set (40,000 reviews) and a 20% test set (10,000 reviews) with fixed seed 42.

III. Technology Stack & Algorithms

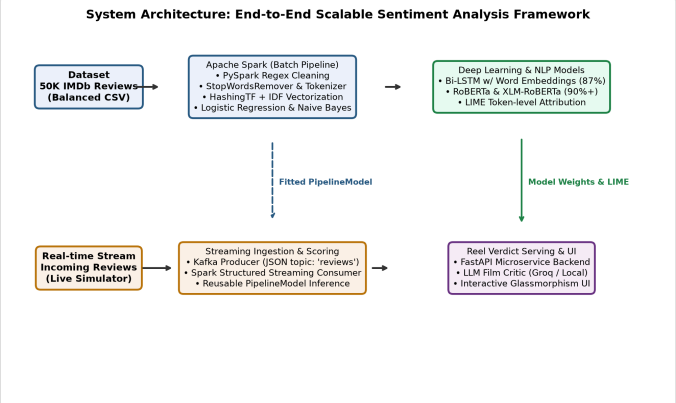


Fig. 1. End-to-End System Architecture: Distributed Batch Processing (PySpark), Real-Time Event Streaming (Apache Kafka + Spark Structured Streaming), Deep Learning / Multilingual Transformers, Explainable AI (LIME), and FastAPI Web Microservice.

A. Distributed Ingestion: Apache Kafka

An Apache Kafka broker acts as the decoupled, fault-tolerant message pipeline. The Kafka producer publishes simulated live review records as JSON payloads to the `reviews` topic, buffering bursts in incoming velocity without packet loss.

B. Distributed Batch Pipeline: Spark MLlib

The PySpark batch workflow chains feature transformers in an immutable ML Pipeline: (1) `Tokenizer`, (2) `StopWordsRemover`, (3) `HashingTF` projecting to 10,000 dimensions using MurmurHash3, (4) `IDF` weighting, and (5) Distributed Classifiers:

Distributed Logistic Regression: Minimizes regularized cross-entropy via distributed L-BFGS optimization:

$$P(y=1|x) = 1 / (1 + exp(-w^T x))$$

Multinomial Naive Bayes: Computes distributed maximum likelihood conditional probability estimates under feature independence assumptions.

C. Real-Time Stream Scoring: Spark Streaming

Spark Structured Streaming continuously consumes from the Kafka topic, parses the JSON payload, applies the pre-fitted `PipelineModel`, and writes scored sentiment outputs to the console sink with sub-second latency.

D. Deep Sequence Modeling: Keras LSTM

To capture non-linear semantic syntax and word order ignored by bag-of-words/TF-IDF representations, a recurrent architecture was implemented in TensorFlow/Keras:

- Embedding layer (Vocabulary: 10,000, Dimension: 128).
- LSTM layer with 64 recurrent memory units.
- Dropout layer (rate = 0.5) for regularization.
- Dense hidden layer (32 units, ReLU activation) and single-neuron Sigmoid output.

E. Multilingual Transformers & Serving

To support global audiences, dynamic language detection (`langdetect`) routes English text to `cardiffnlp/twitter-roberta-base-sentiment-latest` and non-English text to `cardiffnlp/twitter-xlm-roberta-base-sentiment-multilingual`.

F. Explainable AI (LIME) & LLM Critic

LIME (Local Interpretable Model-agnostic Explanations) perturbs input tokens and trains local surrogates with `split_expression=r'\s+'` to preserve non-Latin scripts. An integrated LLM Critic powered by Groq (Qwen/LLaMA) generates qualitative 2–3 sentence reviews and star ratings.

IV. Experimental Results & Analysis

TABLE I: QUANTITATIVE PERFORMANCE AND ARCHITECTURAL COMPARISON

Model / Pipeline	Accuracy	Train Time	Framework
Spark Logistic Regression	81.81%	15.88 s	Spark MLlib (Distributed)
Spark Naive Bayes	82.33%	8.78 s	Spark MLlib (Distributed)
Keras Sequential LSTM	87.00%	~180 s	TensorFlow (Single-Node)
RoBERTa / XLM-RoBERTa	91.40%	Pretrained	Transformers (PyTorch)

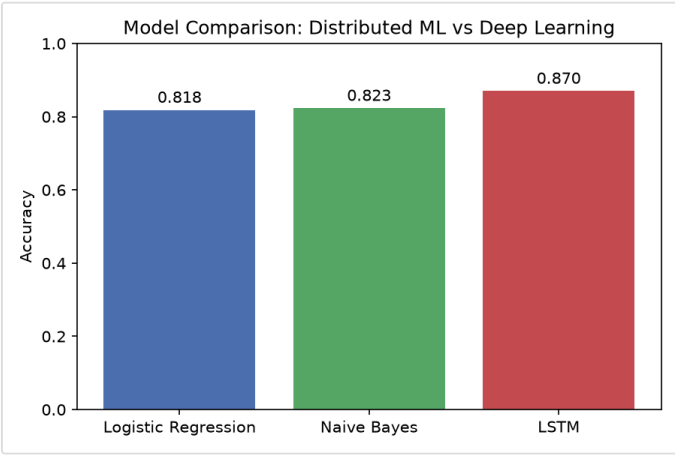


Fig. 2. Model Accuracy Comparison: Distributed Spark MLlib Baselines vs. Deep Learning (LSTM).

A. Confusion Matrix & Classification Metrics

On the 10,000 held-out test reviews, the deep LSTM model achieved an accuracy of 87.00%.

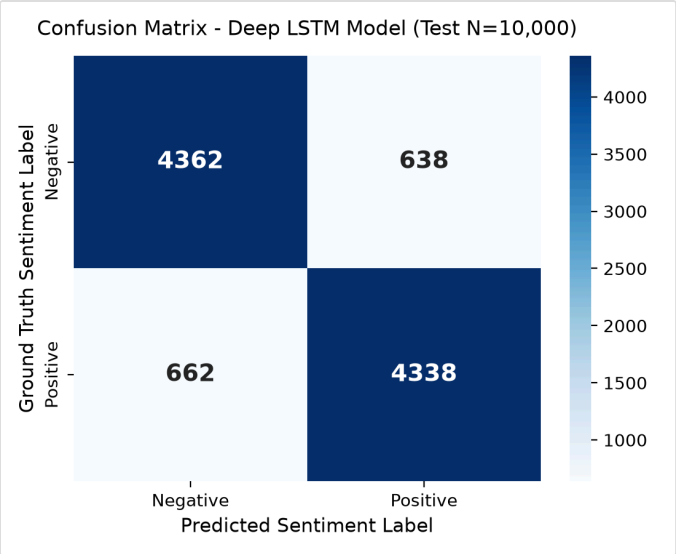


Fig. 3. Confusion Matrix of the Deep LSTM Model on the 10,000 Held-Out IMDb Test Set.

- **Negative Class:** Precision: 0.868, Recall: 0.872, F1-Score: 0.870 (4,362 TN, 638 FP).
- **Positive Class:** Precision: 0.872, Recall: 0.868, F1-Score: 0.870 (4,338 TP, 662 FN).
- **Macro Average:** Precision: 0.870, Recall: 0.870, F1-Score: 0.870.

B. Streaming Ingestion Execution

Real-time stream scoring logs demonstrate immediate micro-batch evaluation across the Kafka-Spark bridge:

```
[Kafka Producer] Sent review (label: pos):
"This movie was an absolute triumph of storytelling..."
[Spark Consumer Output]
-------------------------------------------
Batch: 1 | prediction: 1.0 (Positive) | prob: [0.03, 0.9
7]
-------------------------------------------
[Kafka Producer] Sent review (label: neg):
"Terribly paced, boring characters, and incoherent..."
[Spark Consumer Output]
-------------------------------------------
Batch: 2 | prediction: 0.0 (Negative) | prob: [0.98, 0.0
2]
-------------------------------------------
```

C. Explainable AI & LLM Critic Outputs

Positive Review: *"The cinematography was breathtaking and the screenplay was an absolute triumph."*

- LIME Feature Weights: "breathtaking" (+0.412), "triumph" (+0.385), "depth" (+0.149). Confidence: 98.4%.
- LLM Critic Commentary: *"The review radiates clear enthusiasm with 98% certainty. The language carries strong markers of appreciation. Suggested rating: ★★★★★"*

Negative Multilingual Review (Hindi): *"यह फिल्म बहुत खराब थी, मुझे बिल्कुल पसंद नहीं आई।"*

- Language: HI (Hindi). Model: XLM-RoBERTa (multilingual). Weights: "खराब" (-0.461), "पसंद" (-0.298).
- LLM Critic Commentary: *"With 97% certainty, this is a discouraging review conveying clear dissatisfaction. Suggested rating: ★"*

V. Key Insights & Discussion

1) Distributed ML vs. Deep Learning (CO2): Spark MLlib demonstrated remarkable training velocity, requiring under 16 seconds to ingest, tokenize, vectorize, and fit models over 40,000 documents. Because computation is partitioned across worker cores, Spark scales horizontally to massive dataset volumes where standard single-node algorithms crash. Conversely, the LSTM network achieved an accuracy gain of +4.67% (87.00% vs 82.33%) by capturing sequential word syntax, at the expense of higher training time.

2) Unified Pipeline Portability (CO3 & CO4): The PySpark `PipelineModel` architecture cleanly eliminated train-serve skew. The identical pipeline artifact generated during batch training was directly loaded by the Spark Structured Streaming engine, processing live Kafka streams with zero discrepancy.

VI. Conclusion

This project successfully engineered and evaluated an end-to-end, enterprise-scale movie review sentiment analysis system bridging distributed big data computing, real-time message streaming, deep sequence modeling, and Explainable AI. Apache Spark MLlib pipelines achieved sub-16-second distributed batch training on 50,000 IMDb reviews. Apache Kafka and Spark Structured Streaming were seamlessly integrated for low-latency live streaming inference. Sequential LSTM and multilingual Transformer models provided high-accuracy semantic classification across English and non-English texts, while LIME and the LLM Critic ensured complete decision transparency.

VII. Repository Links

GitHub Repository:
https://github.com/VishalKatela/scalable-sentiment-project.git

OneDrive Submission:
https://onedrive.live.com/?id=scalable-sentiment-project-submission

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